

SKILLWEEK9

TASK1: To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

```
jhansi_priya@Jhansi-Priya: ~/ × + ~
jhansi_priya@Jhansi-Priya:~$ mkdir Skill9
jhansi_priya@Jhansi-Priya:~$ cd Skill9
jhansi_priya@Jhansi-Priya:~/Skill9$ sleep 100 &
[1] 671
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Running                  sleep 100 &
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs -l
[1]+  671 Running              sleep 100 &
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs -p
671
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs -x echo %1
671
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Running                  sleep 100 &
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Running                  sleep 100 &
jhansi_priya@Jhansi-Priya:~/Skill9$ bg %1
-bash: bg: job 1 already in background
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Done                     sleep 100
jhansi_priya@Jhansi-Priya:~/Skill9$ fg %1
-bash: fg: %1: no such job
jhansi_priya@Jhansi-Priya:~/Skill9$ ^C
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
jhansi_priya@Jhansi-Priya:~/Skill9$ |
```

TASK2: To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```
jhansi_priya@Jhansi-Priya:~/Skill9$ sleep 200
^Z
[1]+  Stopped                  sleep 200
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Stopped                  sleep 200
jhansi_priya@Jhansi-Priya:~/Skill9$ bg %1
[1]+ sleep 200 &
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Running                  sleep 200 &
jhansi_priya@Jhansi-Priya:~/Skill9$ fg %1
sleep 200
^C
jhansi_priya@Jhansi-Priya:~/Skill9$ |
```

TASK3: To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

```
GNU nano 7.2 signal.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig)
{
    printf("\nSignal received: %d\n", sig);
}

int main()
{
    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);

    printf("Process ID: %d\n", getpid());
    printf("Press Ctrl+C to send SIGINT\n");

    while (1)
    {
        printf("Running...\n");
        sleep(2);
    }

    return 0;
}
```

[illegible]

```

Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...

Signal received: 15
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...

Signal received: 15
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Killed
jhansi_priya@Jhansi-Priya: ~/Skill9$ |

```

```

jhansi_priya@Jhansi-Priya: ~/ × + ▾
jhansi_priya@Jhansi-Priya:~$ cd Skill9
jhansi_priya@Jhansi-Priya:~/Skill9$ ps
  PID TTY          TIME CMD
  738 pts/2    00:00:00 bash
  752 pts/2    00:00:00 ps
jhansi_priya@Jhansi-Priya:~/Skill9$ ps aux | grep signal
root      298  0.0  0.3 112760 25300 ?        Ssl  05:23   0:
00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
grade-shutdown --wait-for=signal
jhansi_+  727  0.0  0.0   2692  1800 pts/0    S+   05:35   0:
00 ./signal
jhansi_+  754  0.0  0.0   4100  2164 pts/2    S+   05:36   0:
00 grep --color=auto signal
jhansi_priya@Jhansi-Priya:~/Skill9$ kill -TERM 727
jhansi_priya@Jhansi-Priya:~/Skill9$ kill -15 727
jhansi_priya@Jhansi-Priya:~/Skill9$ kill -9 727
jhansi_priya@Jhansi-Priya:~/Skill9$ kill -l
 1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL
 5) SIGTRAP      6) SIGABRT     7) SIGBUS      8) SIGFPE      9) SIGKILL     10)
SIGUSR1
11) SIGSEGV     12) SIGUSR2    13) SIGPIPE     14) SIGALRM     15)
SIGTERM
16) SIGSTKFLT   17) SIGCHLD   18) SIGCONT     19) SIGSTOP     20)
SIGTSTP
21) SIGTTIN     22) SIGTTOU   23) SIGURG      24) SIGXCPU     25)
SIGXFSZ
26) SIGVTALRM   27) SIGPROF   28) SIGWINCH    29) SIGIO       30)
SIGPWR
31) SIGSYS      32) SIGRTMIN  33) SIGRTMIN+1  34) SIGRTMIN+2  35)
SIGRTMIN+3
36) SIGRTMIN+4  37) SIGRTMIN+5  38) SIGRTMIN+6  39) SIGRTMIN+7  40)
SIGRTMIN+8

```

TASK4: To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

```
jhansi_priya@Jhansi-Priya:~/Skill9$ sleep 300 &
[1] 775
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Running                  sleep 300 &
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs -l
[1]+  775 Running                  sleep 300 &
jhansi_priya@Jhansi-Priya:~/Skill9$ kill -STOP 775

[1]+  Stopped                  sleep 300
jhansi_priya@Jhansi-Priya:~/Skill9$ kill -CONT 775
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]+  Running                  sleep 300 &
jhansi_priya@Jhansi-Priya:~/Skill9$ sleep 200
^Z
[2]+  Stopped                  sleep 200
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]-  Running                  sleep 300 &
[2]+  Stopped                  sleep 200
jhansi_priya@Jhansi-Priya:~/Skill9$ bg %1
-bash: bg: job 1 already in background
jhansi_priya@Jhansi-Priya:~/Skill9$ jobs
[1]-  Running                  sleep 300 &
[2]+  Stopped                  sleep 200
jhansi_priya@Jhansi-Priya:~/Skill9$ fg %1
-bash: fg: job has terminated
[1]-  Done                      sleep 300
jhansi_priya@Jhansi-Priya:~/Skill9$ |
```


